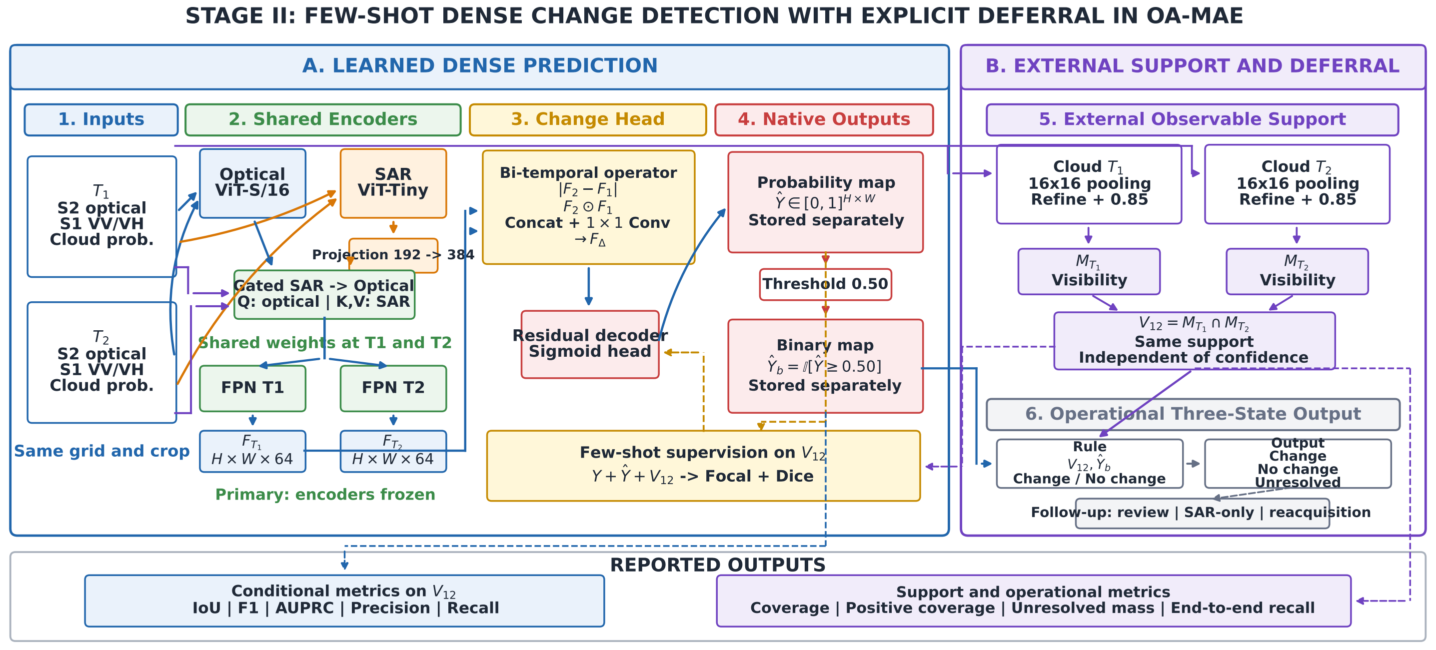

STAGE II: FEW-SHOT DENSE CHANGE DETECTION WITH EXPLICIT DEFERRAL IN OA-MAE
A. LEARNED DENSE PREDICTION
B. EXTERNAL SUPPORT AND DEFERRAL
1. Inputs
2. Shared Encoders
3. Change Head
4. Native Outputs
5. External Observable Support
T_1 S2 optical S1 VV/VH Cloud prob.
T_2 S2 optical S1 VV/VH Cloud prob.
Same grid and crop
Optical ViT-S/16
SAR ViT-Tiny
Projection 192 -> 384
Gated SAR -> Optical Q: optical | K,V: SAR
Shared weights at T1 and T2
FPN T1
FPN T2
F_{T_1} $H \times W \times 64$
F_{T_2} $H \times W \times 64$
Primary: encoders frozen
Bi-temporal operator $|F_2 - F_1|$ $F_2 \odot F_1$ Concat + 1×1 Conv $\to F_\Delta$
Residual decoder Sigmoid head
Few-shot supervision on V_{12}
$Y + \hat{Y} + V_{12}$ -> Focal + Dice
Probability map $\hat{Y} \in [0,1]^{H \times W}$ Stored separately
Threshold 0.50
Binary map $\hat{Y}_b = \mathbb{I}[\hat{Y} \geq 0.50]$ Stored separately
Cloud T_1 16x16 pooling Refine + 0.85
Cloud T_2 16x16 pooling Refine + 0.85
M_{T_1} Visibility
M_{T_2} Visibility
$V_{12} = M_{T_1} \cap M_{T_2}$ Same support Independent of confidence
6. Operational Three-State Output
Rule $V_{12}, \hat{Y}_b$ Change / No change
Output Change No change Unresolved
Follow-up: review | SAR-only | reacquisition
REPORTED OUTPUTS
Conditional metrics on V_{12} IoU | F1 | AUPRC | Precision | Recall
Support and operational metrics Coverage | Positive coverage | Unresolved mass | End-to-end recall